Universal Journal of Green Sci-Tech and Management, Volume 1, Issue 1 (July-August) 2025: Pages 31-37

UNIVERSAL JOURNAL OF GREEN SCI-TECH & MANAGEMENT

Green Sci-Tech & Mangement (UJGSM)

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ISSN: XXXX-XXXX (Online)
www.uorapublications.com

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Performance and Emission Characteristics of a Four-Stroke Engine Using E20 Fuel Blend

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Abstract: *The development of more eco-friendly fuels has promoted the major research of the ethanol-gasoline mixtures. Whereas most of the previous research highlights on E85, the viability of lower ethanol blends like E20 is vital to developing countries because of their cost, availability and compatibility of materials. The experiment assesses the performance and emission features of a four-stroke, air-cooled, single-cylinder spark ignition (SI) engine fuelled by E20 (20% ethanol, 80% gasoline). Key parameters, which were compared using experimental trials, comprised brake thermal efficiency, brake specific fuel consumption (BSFC), exhaust gas temperature, and controlled emissions (CO, HC, NOx) of pure gasoline and E20. The results demonstrate that E20 contributes to a slight gain in thermal efficiency and a drastic decrease in CO and HC emissions and slight progress in NOx formation. It implies that E20 can be used as a non-polluting, cost-effective, and intermediate alternative to pure gasoline in small and medium-size operations.*

Keywords: *Ethanol blend, E20, Spark ignition engine, Emissions, Brake thermal efficiency.*

1. INTRODUCTION

The energy landscape in the world is changing dramatically as the world grows more worried about the exhaustion of fossil fuel, the worsening of the environment, and the necessity of finding alternatives that are sustainable [1]. The traditional petroleum-based fuels though having served as the backbone of transportation over the past 100 years, are receiving growing criticism due to their role in the greenhouse effect, air pollution as well as reliance on crude oil imports. As a result, scholars and policy makers have shifted their focus towards renewable fuels and environmentally friendly fuels, and ethanol has been recognized as one of the best alternatives to this [2-3].

Ethanol is an oxygenated bio-fuel that can be derived out of renewable materials like sugarcane, corn, and lignocelluloses biomass. It is defined by the fact that it produces less carbon monoxide (CO) and unburnt hydrocarbon (HC) emissions than gasoline, has a higher octane rating, and it burns in a cleaner way. The use of ethanol-gasoline blends are already in use in some countries, up to E85 (85% ethanol, 15% gasoline), depending on the infrastructure and compatibility with vehicles [4]. In India, the Government of India has started the Ethanol Blended Petrol (EBP) scheme, that specifies slow steps towards greater ethanol blending with the aim of reaching 20% blending (E20) by 2030 [5].

E20 (20% ethanol, 80% gasoline) is one of the options that are especially noteworthy because it is the compromise between the positive effects on the environment and the low level of changes to be made to the engine. E20, in contrast to higher ethanol levels, does not require major modifications to engine design, fuel dispensation systems, or materials, and thus it can easily be fitted into existing four-stroke spark ignition (SI) engines, widely used in motorcycles, passenger cars, and agricultural machinery, currently [6].

Combination of ethanol and gasoline in four-stroke SI engines has demonstrated a promising outcome concerning the stability in combustion, minimized emissions, and enhanced thermal efficiency. Yet, the issues of reduced calorific value, elevated volumetric fuel consumption, and possible NOx increase are still problematic. Thus, E20 in real-life practice needs to be studied through systematic experimental studies to determine its viability [7].

The current study will examine the performance and emission properties of a single-cylinder four-stroke SI engine running on E20 fuel blend relative to pure gasoline. Such parameters as brake thermal efficiency (BTE), brake specific fuel consumption (BSFC), exhaust gas temperature, and controlled emissions (CO, HC, NOx) are considered. The research is part of the realization of the E20 as a transitional green fuel, which is the aim of the world toward sustainability, and at the same time, the small-scale engines can be economically and technically viable [8-14].

2. LITERATURE REVIEW

Extensive studies have already been done on the implementation of ethanol gasoline blends with a significant portion of it, E20 (20% ethanol, 80% gasoline or diesel); this strategy is seen as a feasible measure to minimize emissions and partially substitute fossil fuels. This section cites important research findings touching upon engine performance, emission behavior, and engine adaptation needs when the engine cut on E20 blends.

2.1 Engine Performance with E20

A number of studies have shown that E20 blends typically have a slight beneficial effect of Brake Thermal Efficiency (BTE) over conventional fuels. This increase, which is generally 2-5 percent, is pointed out to the extra oxygen atom in ethanol, which enhances a further complete combustion [1-4]. Nevertheless, the reduced calorific value of ethanol causes a minor rise in the Brake Specific Fuel Consumption (BSFC), typically by 3-5 percent [1-3].

Compared to engine power and torque, it has mixed results. Although a few studies have found that the power output may decline by 25 percent when it is operated on E20 [13, 5], others suggest a slight rise in torque (by about 2 percent) and power (by an average of 23), under specific operating conditions, especially in spark-ignition engines [4, 5]. These results indicate that intermediate ethanol ratios, like E20, have the best performance results, and that an increase in ethanol ratios can have a negative influence on efficiency.

2.2 Emission Characteristics

Among the most promising benefits of the ethanol blends, one can indicate the reduction of the Carbon Monoxide (CO) and Hydrocarbon (HC) emissions. It has been demonstrated through multiple studies that E20 has the ability to record 30-50% of CO and HC emissions as opposed to the pure gasoline or diesel [1,3,4,6,7]. This is improved because ethanol is oxygenated in nature and promotes full combustion.

Conversely, Nitrogen Oxides (NOx) and Carbon Dioxide (CO_2) emissions are more likely to be slightly higher (by around 510 per cent) with E20, especially at higher loads and velocities. This upward trend is associated with the elevated temperatures of combustion and oxygen-enriched fuel which supports the development of NOx [13,8,19].

A downward trend is also evident with the particulate emissions. Research has shown that ethanol-gasoline mixtures reduce the smoke quality and particle count emissions particularly during lean-burn operation [9,10]. This renders E20 a good option in the context of bettering the air.

2.3 Engine Alterations and Operating Condition.

One of the biggest strengths of E20 is that it does not normally necessitate other major engine changes, as compared to higher levels of ethanol like E85. E20 is proposed to be utilized in most engines operating on spark-ignition with only minor modifications [5,7]. Moreover, research on partially insulated (low heat rejection) engines suggests that the combination of E20 and thermal insulation plans can enhance the benefit of the emission, and that part-load benefits can reach as much as 50% CO and 48% total hydrocarbons (THC) in this case [7].

2.4 Comparative Summary

A synthesis of the reviewed studies is presented in Table 1, summarizing the typical effects of E20 on performance and emissions relative to pure fuels.

Table 1: Comparative Effects of E20 Blend on Engine Performance and Emissions

Parameter	Effect of E20 vs. Pure Fuel	Citations
Brake Thermal Efficiency	↑ 2–5%	[1–4]
Engine Power	↓ 2–5% (some ↑ 2–3%)	[1–3,5]
BSFC	↑ 3–5%	[1–3]
CO/HC Emissions	↓ 30–50%	[1,3,4,6,7]
NOx/CO_2 Emissions	↑ 5–10%	[1–3,8]
Particulate Emissions	↓ significantly	[9,10]

3. METHODOLOGY

This section outlines the experimental setup, test conditions, and procedures adopted for evaluating the performance and emission characteristics of a four-stroke spark ignition (SI) engine fueled with gasoline and E20 blend (20% ethanol + 80% gasoline).

3.1 Engine Specifications

The experiments were conducted on a **single-cylinder, four-stroke, air-cooled SI engine**, commonly used in motorcycles and small passenger vehicles. The detailed specifications are presented in Table 1.

Table 1: Engine Specifications

Parameter	Specification
Engine Type	Four-stroke, SI engine
Cylinder	Single
Cooling System	Air-cooled
Displacement	150 cc
Compression Ratio	9.2 : 1
Rated Power	11 HP @ 7500 RPM
Fuel Used	Gasoline, E20 (20:80)

3.2 Fuel Properties

The selected test fuels were **pure gasoline (E0)** and **E20 blend**. The blend was prepared by mixing 20% anhydrous ethanol with 80% gasoline by volume. Table 2 presents the key properties of the fuels.

Table 2: Properties of Test Fuels

Property	Gasoline (E0)	E20 Blend
Density (kg/m³)	740	755
Calorific Value (MJ/kg)	44.0	40.5
Octane Number (RON)	91	95–96
Oxygen Content (%)	~0	~7

3.3 Experimental Setup

The engine was mounted on a **computerized engine test rig** equipped with:

a) Eddy current dynamometer for measuring brake power and torque.
b) Exhaust gas analyzer for measuring CO, HC, and NOx emissions.
c) Thermocouples (K-type) for monitoring exhaust gas temperature.
d) Fuel burette method for accurate measurement of fuel consumption.

The entire system was interfaced with a **data acquisition system** for continuous recording of performance parameters.

3.4 Test Procedure

a) The engine was initially operated on **pure gasoline (E0)** to obtain baseline performance data.
b) The same tests were repeated with **E20 fuel blend**, maintaining identical load and speed conditions.
c) The operating range covered **2000 to 7000 RPM**, with data recorded at regular intervals.
d) For each fuel type, the following parameters were measured:

   1. Brake Thermal Efficiency (BTE)
   2. Brake Specific Fuel Consumption (BSFC)
   3. Exhaust Gas Temperature (EGT)
   4. Emission concentrations: CO, HC, and NOx

e) Each test was conducted three times to minimize experimental error, and average values were considered for analysis.

3.5 Performance and Emission Calculations

- **Brake Power (BP):** Calculated from dynamometer readings.
- **Brake Thermal Efficiency (BTE):**

$$BTE = \frac{BP}{Fuel\ Energy\ Input\ Rate}$$

- **Brake Specific Fuel Consumption (BSFC):**

$$E = \frac{Fuel\ Mass\ Flow\ Rate}{BP}$$

- Emissions were directly obtained from the exhaust gas analyzer and compared between the two fuels.

4. Results and Discussion

The experimental results obtained with **gasoline (E0)** and **E20 fuel blend** were analyzed for performance and emissions. The observations are summarized below.

4.1 Brake Thermal Efficiency (BTE)

E20 showed 4-6 percent betterment in BTE than gasoline. This enhancement is credited to the fact that ethanol increases the amount of oxygen in the air and this increases the level of combustion.

Table 3: Brake Thermal Efficiency

Fuel	BTE (%)	% Change
Gasoline (E0)	26.5	–
E20 Blend	28.0	+5.7%

4.2 Brake Specific Fuel Consumption (BSFC)

The BSFC was slightly higher for E20 (approx. 5%) due to ethanol's lower calorific value, which requires greater fuel flow to generate equivalent power.

Table 4: Brake Specific Fuel Consumption

Fuel	BSFC (kg/kWh)	% Change
Gasoline (E0)	0.32	–
E20 Blend	0.34	+6.3%

4.3 Exhaust Gas Temperature (EGT)

The EGT was lower in E20 operation compared to gasoline. Ethanol's higher latent heat of vaporization absorbs more heat during vaporization, reducing combustion chamber temperature and exhaust heat.

Table 5: Exhaust Gas Temperature

Fuel	EGT (°C)	% Change
Gasoline (E0)	420	–
E20 Blend	395	–6.0%

4.4 Emission Characteristics

1. Carbon Monoxide (CO): Reduced by ~24% with E20.
2. Hydrocarbons (HC): Reduced by ~22% due to better flame propagation.
3. Nitrogen Oxides (NOx): Slightly increased (~8%) with E20 because of higher combustion temperatures and oxygen availability.

Table 6: Emission Results

Emission Parameter	Gasoline (E0)	E20 Blend	% Change
CO (% vol)	2.1	1.6	–24%
HC (ppm)	820	640	–22%
NOx (ppm)	710	770	+8%

4.5 Overall Discussion

The results confirm that E20 improves combustion efficiency, lowers CO and HC emissions, and reduces exhaust gas temperature, making it a cleaner fuel alternative. Although fuel consumption and NOx emissions are slightly higher, these effects are manageable and do not outweigh the benefits.

Thus, E20 is a practical and eco-friendly alternative to gasoline for four-stroke SI engines without requiring significant engine modifications.

Table 7: Comparative Results of Gasoline vs. E20 in Four-Stroke Engine

Parameter	Gasoline (E0)	E20 Blend	% Change
Brake Thermal Efficiency (BTE, %)	26.5	28.0	+5.7%
Brake Specific Fuel Consumption (BSFC, kg/kWh)	0.32	0.34	+6.3%
Exhaust Gas Temperature (EGT, °C)	420	395	–6.0%
Carbon Monoxide (CO, % vol)	2.1	1.6	–24%
Hydrocarbons (HC, ppm)	820	640	–22%
Nitrogen Oxides (NOx, ppm)	710	770	+8%

5. CONCLUSION

The experimental study, which was conducted on a four-stroke single-cylinder spark ignition engine using the E20 blend (20% ethanol and 80% gasoline) as fuel, shows that ethanol can be considered as a good alternative fuel that can be used in small-scale applications. In a nutshell, the findings can be stated to be as follows:

1. Performance: E20 recorded an increase of 5.7 percent of Brake Thermal Efficiency (BTE) over gasoline, which verified an improvement in combustion efficiency. There was a slight rise in Brake Specific Fuel Consumption (BSFC, +6.3) as a result of low calorific value of ethanol.
2. Combustion Characteristics: With E20, the Exhaust Gas Temperature (EGT) dropped by 6.0% which showed that combustion was cooler and could benefit engine durability.
3. Emissions: E20 operation caused high Carbon Monoxide (24) and Hydrocarbons (22) to be reduced thus demonstrating that it is a cleaner burner. Nevertheless, Nitrogen Oxides (NOx) went up slightly (+8) owing to increased oxygen levels and increased combustion temperature.
4. In general, the research provides a positive result, as it demonstrates that E20 presents a fair compromise between the performance increase, decreased emission, and operational viability. The mixture is compatible with current four stroke SI engines with only slight modification and thus is a feasible option in the current ethanol blending programs and sustainable fuel transitions.

Future Scope: Long term engine life, optimization of ignition timing and compatibility of materials with the ethanol blends should be further studied as well as practical field tests of the motorcycles and small passenger vehicles.

Conflicts of Interest:

The authors declare no conflicts of interest about the publication of this research paper.

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